

DBMS Theory (all in one)

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Question Format

MCQ

- 1. functional dependency, armstrongs axiom and one proof**
- 2. relational algebra**
- 3. random question**
- 4. Normalization**
- 5. EER to relational conversion**
- 6. EER diagram**

1. which type of entity cannot exist in the database unless another type of entity also exists in the database but does not exist require that

the identifier of that other entity be included as part of its own identifier(2024,2022)

- a. weak entity.**
- b. strong entity.**
- c. ID-dependent entity.**
- d. ID-independent entity.**

answer: a

2. The different classes of relations created by the technique for preventing modification anomalies are called:(2024,2022,2018)

- a. Normal forms.**
- b. Referential integrity constraints.**
- c. functional dependencies.**
- d. None of above is correct.**

answer: a

3. A keys:(2024)

- a. Must always be composed of two or more columns.**
- b. can only be one column.**
- c. identifies a row.**
- d. identifies a column.**

answer: c

4. If a table has been normalized so that all determinants are candidate keys, then the table is in:(2024)

- a. 1NF.
- b. 2NF.
- c. 3NF.
- d. BCNF.

answer: d

5. When the values is one or more attributes being used as foreign key must exist in another set of one or

more attributes in another tables, we have created a(n):(2024,2022)

- a. transitive dependency.
- b. insertion anomaly.
- c. referential integrity constraint.
- d. normal form.

answer: c

6. A functional dependency is a relationship between or among:(2024)

- a. tables.

- b. rows.**
- c. relations.**
- d. attributes.**

answer:d

7._____ is the minimal superkey.(2022)

- a.primary key**
- b.candidate key**
- c.foreign key**
- d.alternate key**
- e.unique key**

answer:b

8.Metadata about structure of database is stored in:(2022)

- a.data files**
- b.indices**
- c.data dictionary**
- d.metadata files**

answer:c

9.In the relational model ,relationship between relations or tables are created by:(2022,2018)

- a.composite key**
- b.determinants**
- c.candidate keys**
- d.foreign keys**

answer:d

10.A combination of Cartesian Product followed by selection process:(2022)

- a.association**
- b.product formalization**
- c.joins**
- d.protocol**

answer:c

11.Functional dependencies are the types of constraints that are based on:(2018)

- a.keys**
- b.key revisited**
- c.superset keys**
- d.none of these**

answer:a

12.Database ----- is the logical design of the data and the database -----

which is snapshot of data at a given instance of time.(2018,2023)

a.instance, schema

b.relation, schema

c.relation, domain

d.schema, instance

answer:d

13.The tuples of the relations can be of _____ order.(2018,2023)

a.any

b.same

c.sorted

d.constant

answer:a

14.Which one of the following is a set of one or more attributes taken collectively to uniquely

identify a record:(2018,2023)

a.candidate key

b.sub key

- c.super key**
- d.foreign key**

answer:c

15.Tables in Second Normal Form:(2NF) (2018,2023)

- a.eliminates all hidden dependencies**
- b.eliminates all possibilities of insertion anamolies**
- c.have a composite key**
- d.have all non-key fields depend on the whole primary key**

answer:d

16.For each attribute of a relation, there is a set permitted values, called the_____ of that attribute. (2023)

- a.Domain**
- b.Relation**
- c.Set**
- d.Schema**

answer:a

17. An attribute in an relation is a foreign key if the _____ key from one relation is used as an attribute in that relation.(2023)

a.Candidate

b.Super

c.Primary

d.Sub

answer:c

18.Which forms simplifies and ensures that there is minimal data aggregates and repetitive groups:(2023)

a.1NF

b.2NF

c.3NF

d.All of the mentioned

answer:c

19.Several entities are associated through (2023)

a.connection

b.relationship

c.agreement

d.integration

answer:b

20. A collection of tables to represent data and relationship among data is represented through model----- (2023)

a. ER model

b. Relational model

c. semistructured model

d. object based model

answer: b

1. Define function dependency!

List the Armstrong axioms and state the below statement

$\{WX \rightarrow Y, X \rightarrow Z, Z \rightarrow WY\} \models \{X \rightarrow Y\}$

ANSWER:

A functional dependency $x \rightarrow y$, between two sets of attributes specifies a constraint that any two tuples t_1 and t_2 that have $t_1[x] = t_2[x]$, they must also have $t_1[y] = t_2[y]$

Armstrong's Axioms:

1. The Reflexive Rule: If y is a proper subset of x , then $x \rightarrow y$

2. The Augmentation Rule: $\{x \rightarrow y\} \models xz \rightarrow yz$

3. The Transitivity Rule: $\{x \rightarrow y, y \rightarrow z\} \models x \rightarrow z$

demonstration:

1. $X \rightarrow Z$

2. $Z \rightarrow WY$

3. $x \rightarrow wy$ (transitivity)

4. y is a subset of w . $wy \rightarrow y$ (reflexivity)

5. $\{x \rightarrow wy, wy \rightarrow y\} \models x \rightarrow y$ (transitivity)

Demonstration:

1. $\{X \rightarrow Z\} \models WX \rightarrow WZ$ (augmentation)

2. $\{Z \sqsubseteq WZ\} \models WZ \rightarrow Z$ (reflexivity)

3. $\{WX \rightarrow WZ, WZ \rightarrow Z\} \models WX \rightarrow Z$ (transitivity)

2.solve the question

14.8 The table below lists customer/car hire data. Each customer may hire cars from various outlets throughout Glasgow. A car is registered at a particular outlet and can be hired out to a customer on a given date.

jobID	pickupDateTime	driverID	dFName	dLName	clientID	cFName	cLName	cAddress
1001	25/07/00.10.00	I456	Jane	Watt	C034	Anne	Way	111 Storrie Road, Paisley
1102	29/07/00.10.00	I456	Jane	Watt	C034	Anne	Way	111 Storrie Road, Paisley
1203	30/07/00.11.00	I344	Tom	Jones	C034	Anne	Way	111 Storrie Road, Paisley
1334	2/08/00.13.00	I666	Karen	Black	C089	Mark	Fields	120 Lady Lane, Paisley
1455	2/08/00.13.00	I957	Steven	Smith	C019	John	Brown	13 Renfrew Road, Paisley
1676	25/08/00.10.00	I344	Tom	Jones	C039	Karen	Worth	34 High Street, Paisley

- The data in the table is susceptible to update anomalies. Provide examples of how insertion, deletion, and modification anomalies could occur on this table.
- Identify the functional dependencies represented by the data shown in the table. State any assumptions you make about the data.
- Using the functional dependencies identified in part (b), describe and illustrate the process of normalization by converting Table 1 to Third Normal Form (3NF) relations. Identify the primary and foreign keys in your 3NF relations.

Solution (a):

Insertion Anomalies: If a new driver joins the outlet, then driver cannot be added to database if no rides are done by him

Update Anomalies: If client address is updates then that needs to be chaged in all the tuples.

Deletion Anomalies: If some ride detail is deleted and that is the only ride taken by client or delivered by driver, then driver and client details also get deleted.

Solution (b):

driverID-> dFName, dLName

clientID-> cLName, cFName, cAddress

jobID -> pickupDateTime, driverID, clientID

Solution (c):

Primary Key: jobID

Thus driverID-> dFName, dLName and clientID-> cLName, cFName, cAddress are partial dependencies. Thus tables is not in 2NF.

Table T-1(jobID, pickupDateTime, driverID, clientID)

Table T-2(driverID, dFName, dLName)

Table T-3(clientID, cLName, cFName, cAddress)

Since no transitive dependencies exist, tables T-1, T-2 and T-2 are in 3NF.

3.Given the relational schema:

4 points

Jedi-Tamas (master, apprentice)

jedi(name,side,home-planet)

government(leader , planet, position)

inhabitants(specie, planet)

write an algebraic expression which will query (by listing their names) to find all “wookie” (which is a specie) jedi on the light side of the force!

[for half marks write an sql query]

$$\Pi_{\text{name}}(\sigma_{\text{specie}='wookie'}(\text{inhabitants}) \bowtie_{\text{inhabitants.planet}=\text{jedi.home-planet}} \sigma_{\text{side}='light'}(\text{jedi}))$$

4. From an SQL user's perspective , does relational model provide logical and physical independence?why? 2 points

yes.

the relational model provides both logical and physical independence.

Logical Independence: Users can change the logical schema (e.g., add new fields or tables) without affecting the application programs.

Physical Independence: Users can change the physical storage (e.g., how data is stored or indexed) without affecting the logical schema or application programs.

5.Normalization - Solved exercise

Question:

Consider a relation Movies_Screened with attributes Theatre, Movie, Day, Time, and Certificate. Sample tuples are as follows:

Sathyam, 'Slumdog Millionaire', Wed, 18:00, 15

Sathyam, 'Slumdog Millionaire', Wed, 20:00, 15

PVR, 'Slumdog Millionaire', Wed, 20:30, 15

PVR, 'Vicky Christina Barcelona', Wed, 20:30, 12A

Each movie is assigned a certificate by the Indian Board of Film Certification; the certificate value 15 means that nobody younger than 15 years of age can see this movie in a cinema. The same theatre can show a movie on multiple times during a day, and may show different movies at the same time (on different screens).

(a) Does this relation violate the second normal form requirements? Explain.

(b) Decompose this relation into BCNF, and explain why the resulting relations are in BCNF.

Answer (a):

To check for 2NF, first we need to find the candidate keys for MOVIES_SCREENED.

Let us find the functional dependencies (FDs) of MOVIES_SCREENED.

- THEATRE cannot determine any attributes as a theatre screens more than one movie, it screens on different days, different timings, and different certification movies.
- MOVIE can determine the CERTIFICATE value as a movie will be given only one certificate. Hence, we can include $\text{MOVIE} \rightarrow \text{CERTIFICATE}$.
- Likewise, DAY, TIME and CERTIFICATE cannot determine the other attributes uniquely.

We get the set of FDs for this relation as follows;

$F = \{ \text{MOVIE} \rightarrow \text{CERTIFICATE}, (\text{THEATRE}, \text{MOVIE}, \text{DAY}, \text{TIME}) \rightarrow \text{CERTIFICATE} \}$

To find the candidate key, we need to find the closure of left hand side attributes of the FDs.

$(\text{THEATRE}, \text{MOVIE}, \text{DAY}, \text{TIME})^+ = \text{THEATRE}, \text{MOVIE}, \text{DAY}, \text{TIME}, \text{CERTIFICATE}$.

Hence, the composite key (THEATRE, MOVIE, DAY, TIME) is the candidate key for the relation MOVIES_SCREENED.

To be in 2NF, a relation should not have partial functional dependency.

In our relation, a non-key attribute CERTIFICATE is determined by MOVIE, which is part of a candidate key (THEATRE, MOVIE, DAY, TIME). So the given relation is not in 2NF.

The relation MOVIES_SCREENED violates second normal form.

Answer (b):

As discussed, the relation violates 2NF. To normalize to 2NF, we decompose the relation using the violating functional dependency $MOVIE \rightarrow CERTIFICATE$.

It results in the following relations;

Movie_Screens (THEATRE, MOVIE, DAY, TIME)

Movies (MOVIE, CERTIFICATE).

Both relations are in 2NF because no partial dependency exists [see the keys underlined].

Both relations are in 3NF too because no transitive dependencies found.

Also, both are in BCNF because in the Movie_Screens relation, no subset of the attributes determines any other attribute, and the only non-trivial dependency in MOVIES is from MOVIES to CERTIFICATE.

6.solve the normalization question.

3. The table shown below stores details of students and the overall grade each student obtained in different modules. The Primary Key is (StudentID, ModuleID).

<u>StudentID</u>	<u>StudentName</u>	<u>ModuleID</u>	<u>ModuleName</u>	Grade
S001	Smith	M01	Java	A
S001	Smith	M02	Databases	B
S002	Ford	M01	Java	B

(i) Which Normal Form does the above table violate and why?

(ii) Give an example of an update anomaly and an example of a delete anomaly that may occur if the table is left un-normalised. Explain the problems that are caused.

(iii) Show how you would normalise the table.

((((answer))))

(i) The table violates 2nd Normal Form. because there are two partial dependencies: StudentID → StudentName and ModuleID → ModuleName.

(ii) Update anomaly: if a module is to be renamed, it is possible to mistype one of its instances, thus introducing an inconsistency.

Deletion anomaly: if students are deleted, modules will also be deleted, resulting in a loss of modules' details.

(iii)

Student(StudentID , StudentName)

Module(ModuleID , ModuleName)

Results(StudentID *, ModuleID *, Grade)

7.solve the problem

4. A company uses the table below to record details of staff. Each staff has up to three qualifications:

<u>StaffID</u>	<u>staffname</u>	<u>qualifications</u>
S01	Ibanga	BSc, MSc, PhD
S02	Kumar	BSc, MSc
S03	Grant	BSc, PhD

(i) Explain why this table is not in "First Normal Form" (1NF).

(ii) Show how this table can be transformed into 1NF tables.

answer

(i) The attribute "qualifications" is multi-valued on the PK, hence the table is not in 1NF.

(ii) Solution 1:

Students (StaffID , StaffName, Qualification1, Qualification2, Qualification3)

Solution2:

Students (StaffID , StaffName) (2 marks)

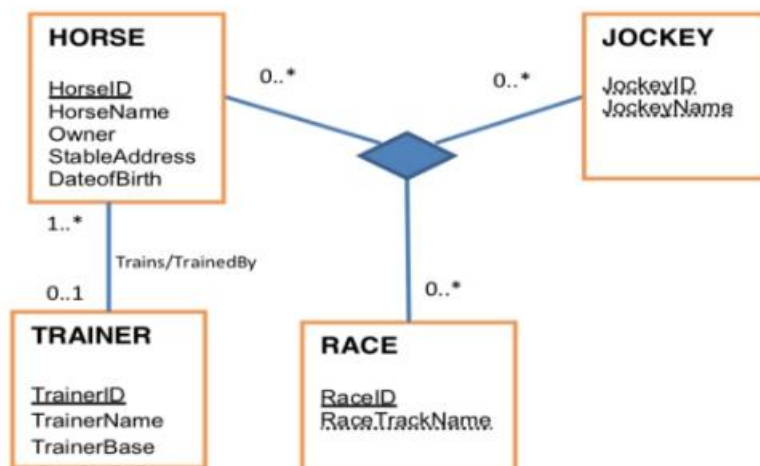
Qualifications (StaffID *, Qualification) (2 marks)

Alternatively, students can use a third table (QualID , Qualification)

5. Refer to the ER model given in Figure A1. Assume that :-

A Trainer x trained by no more than one Trainer.

A given horse competes in a particular race ridden by a particular jockey on a specific date and the position (ie 1st, 2nd etc) in which a horse finishes is recorded.



- a. In the context of database design give a precise definition of an Entity Type

answer

Entity type describes an object, a noun, a class any meaningful collection of objects of the

same type having the same properties/attributes

- b. Explain using examples from the ER model (figure A1) the difference between a ternary relationship and a binary relationship.

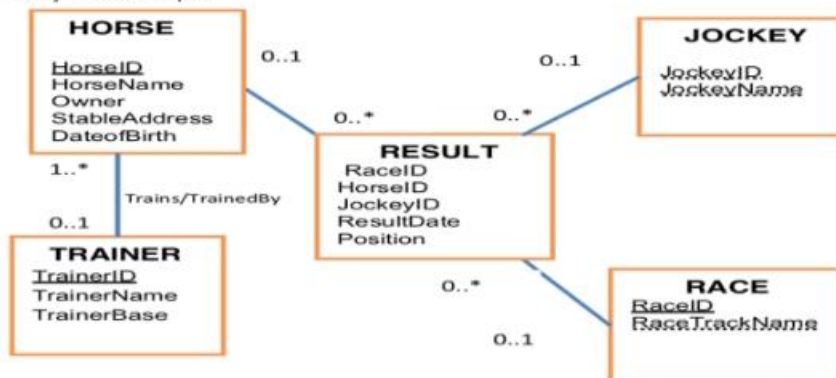
Answer

One mark each for identifying these relationships. Ternary relationships have three participating entity types whereas binary have two. Example : results and trains respectively. An example will show diagrammatically examples of instances of horseID , jockeyID , RaceID all participating in the same relationship. Binary relationship instances have a set of instances that relate a horse to a particular trainer and a trainer related to many horses. Some instances of a horse may not have a (current) trainer.

The accuracy of the cardinality and participation is important in both cases.

- The existence of three relationship instances (s, p), (j, p), and (s, j) does not necessarily imply that an instance (s, j, p) exists in the ternary relationship

c. explain how you would modify the ER model given in Figure A1 in order to resolve many to many relationships.



d. Explain how you would translate your modified ER model (in part d) as a set of Tables, giving table names, column names and an indication of primary and foreign keys.

answer

Many to many relationships automatically introduce an associative entity that maps directly to a new table with a composite key consisting of fields that exist in the original tables forming the relationship. Hence the ternary relationship would be transformed into a table with a composite key comprising of the three identifiers from the three entity types forming the relationship. One to Many relationships map to two tables generally although technically this usually depends on the participation constraint on the one side.

Result Table (note italics/underlined = Foreign Key /bold = Primary Key.

HorseID, JockeyID, ResultDate, RaceID, Position

Horse table

HorseID, horseName, Owner, DateOfBirth, StableAddress, TrainerID

Trainer table

TrainerID, TrainerName, TrainerBase...

Jockey table

jockeyID, jockeyName

8.solve the normalization question:

4. The following table stores details of doctors, patients and dates of appointments. The Primary Key is (doctorID, patientID).

Appointments

doctorID	doctorName	patientID	patientName	date
D01	Kumar	P02	Smith	10-Aug-16
D01	Kumar	P01	Ford	5-Sep-16
D02	Robinson	P02	Smith	10-Aug-16

(2 points)

a. Explain why this table is not in 2nd normal form.

b. Describe two types of anomaly that could be caused by update, insert or delete operations giving an example of each, with reference to the above table. (4 points)

c. Transform the table into 2nd normal form. Show the structures of the resultant tables. (4 points)

- a. there is partial dependencies like doctorID -> doctorName
 b. deletion anomaly: If we delete the appointment of Smith, we will lose the detail of a doctor

update anomaly: if we update patient name, we need to update in two rows otherwise the data will be inconsistent

c. doctor(doctorID, doctorName)

patient(patientID, patientName)

Appointment(doctorID*, patientID*, date)

WAREHOUSE

The warehouse database stores inventory information. A warehouse has one manager who has an employee number (unique) and a name. A warehouse is identified by its warehouse name (unique) and its address. Every part in the warehouse has a part number (unique), inventory date, quantity-on-hand, supplier name (one), and the delivery number, date, and quantity. One delivery could deliver several parts. The data is shown below:

WAREHOUSE (manager-id, manager-name, warehouse-name, warehouse-address, part-no, inventory-date, qty-on-hand, supplier-name, delivery-no, delivery-date, delivery-qty)

manager-id	manager-name	warehouse-name	warehouse-address	part-no	inventory-date	qty-on-hand	supplier-name	delivery-no	delivery-date	delivery-qty
0001	John Duster	South	1001 St Paul St	A001	7/28/2015	5	Vollzo	D001	7/15/2016	5
0001	John Duster	South	1001 St Paul St	A002	7/23/2015	6	Vollzo	D001	7/15/2016	10
	John		1001 St Paul						08-08-	

3. WAREHOUSE Database:

Given details:

.

WAREHOUSE database has below details in it:

- a. Warehouse manager: manager-id, manager-name
- b. warehouse details: warehouse-name, warehouse-address
- c. part details: part_no, inventory-date, quantity-on-hand
- d. supplier details: supplier-name
- e. part-supply details: delivery-no, delivery-date, many parts(delivery-qty, part-no)

.

Functional Dependencies:

.

Below is the list of functional dependencies in the given details:

- a. manager_id -> manager_name
- b. warehouse_name-> warehouse_address, manager_id
- c. part_no -> supplier_name
- d. warehouse_name, part_no -> inventory-date, quantity-on-hand
- e. delivery_no, part_no->delivery-qty, delivery-date

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Normalized tables:

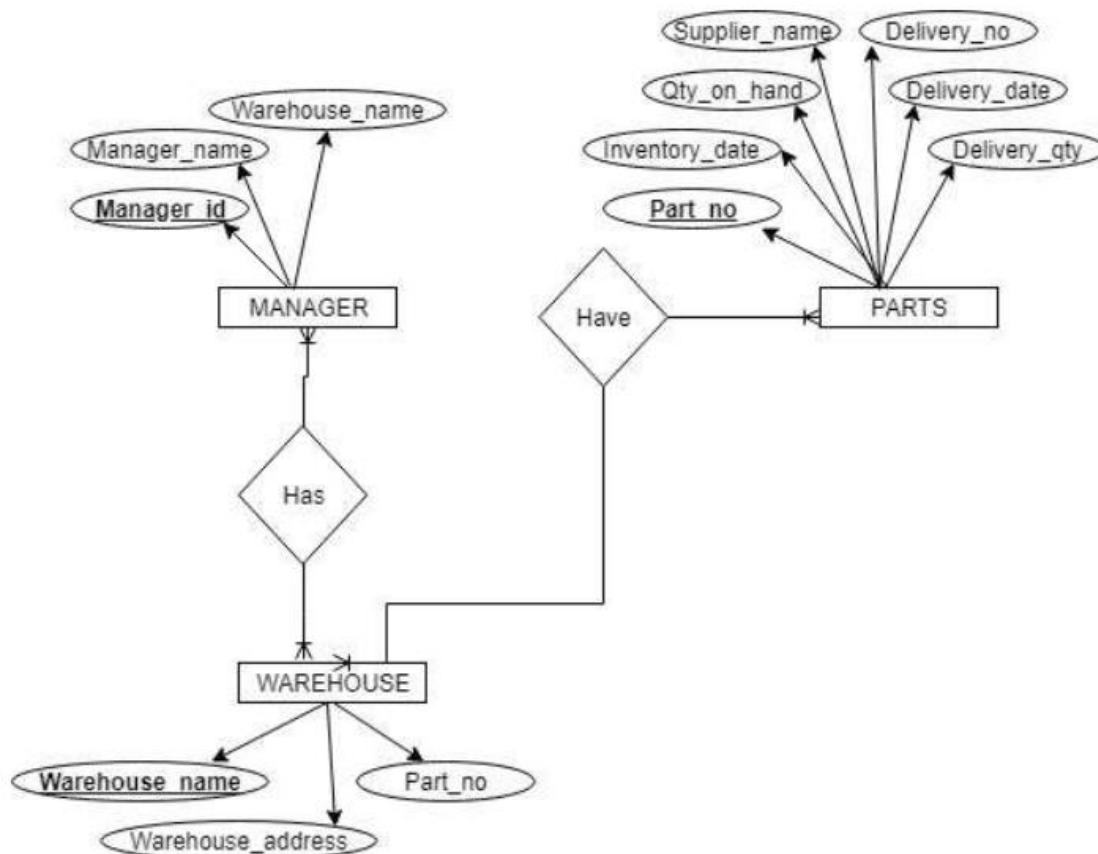
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- a. WarehouseManager(manager_id(fk), manager_name)
- b. Warehouse(warehouse_name(pk), warehouse_address, manager_id(fk))

c. Part(warehouse_name(pk+fk),part_no(pk), inventory_date, quantity_on_hand, supplier_name)

d. Delivery(delivery_no(pk), part_no(pk)(fk), delivery-qty)

ER diagram



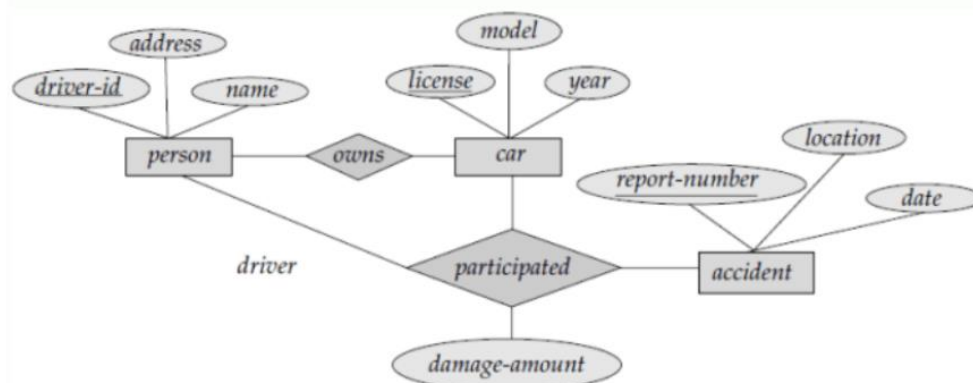
This is asked in 2024 22nd may.

9.solve the ER diagram

3:

(a) Construct an E-R diagram for a car-insurance company whose customers own one or more cars each. Each car has associated with it zero to any number of recorded accidents.

Answer :



E-R diagram for a Car-insurance company.

(b) Construct appropriate tables for the above ER Diagram ?

Car insurance tables:

person (driver-id, name, address)

car (license, year, model)

accident (report-number, date, location)

participated(driver-id, license, report-number, damage-amount)

this was asked several times

10.solve the normalization problem

The table below lists customer/car rental data. Each customer may rent cars from various shops. A car is registered at a particular shop and can be rented out to a customer on a given date.

carReg	make	model	custNo	custName	rentalDate	shopNo	shopLoc
M565 0GD	Ford	Escort	C100	Smith, J	14/5/98	01	Bearsden
M565 0GD	Ford	Escort	C201	Hen, P	15/5/98	01	Bearsden
N734 TPR	Nissan	Sunny	C100	Smith, J	16/5/98	01	Bearsden
M134 BRP	Ford	Escort	C313	Blatt, O	14/5/98	02	Kelvinbridge

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CUSTOMER(custNo (pk), CustName)
CAR(carReg(pk), Make, Model,shopNo)
SHOP(shopNo (pk), shopLoc)
RENTAL(rentalID(PK),rentalDate, custNo, carReg)
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11. CD Database:

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CD database has below details:

CD Details: including ID, Title, Year_Published, Year_purchased

Artist Details: Artist name where the name can repeat for Many CD's, thus need to have Identifier for an artist

Genre Details: Genre details where same genre can have many CD's thus need to have an identifier for Genre

Track Details: Track details including title, length and adding an identifier as it is assumed that various CD's can have same track name(title). track table will be separated.

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Functional Dependencies:

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Below is the list of functional dependencies in the given details:

a. ArtistID-> Name

b. GenreID ->Name

c. ID -> Title, Year_Published, Year_purchased, Artist_id, GenreID

d. TrackID-> title, length, CDID

.

Normalized tables:

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The normalized tables are as below:

a. . Artist(ArtistID(pk), Name)

b. Genre(GenreID(pk), Name)

c. CD(ID(pk), Title, Year_Published, Year_purchased, Artist_id(fk), GenreID(fk))

d. Track(TrackID(pk), title, length, CDID)

.

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12.solve the normalization problem

An embassy records details of interviews of visa applicants in the table below. Interviews are

conducted by members of staff in some of the embassy rooms. In any given day, a member of

staff tends to use the same room throughout that day. An applicant cannot have two interviews in

the same day :

R :

<u>applicantNo</u>	<u>interviewDate</u>	<u>interviewTime</u>	<u>staffNo</u>	<u>roomNo</u>
AP1	13.5.2016	10:30	S5	R101
AP2	13.5.2016	12:00	S5	R101
AP3	13.5.2016	12:00	S9	R200
AP2	22.9.2016	10:30	S5	R200

- List all possible candidate keys for the above relation :

{applicantNo,interviewDate,interviewTime}

{staffNo,interviewDate,interviewTime}

{roomNo,interviewDate,interviewTime}

13. Design an E-R Diagram for a marketplace company (such as e-Bay) where individuals can sell and buy products.

The ER must be able to keep track of customers who sell items and customers who bought them; some customers may do both.

Customer info must include street address, city, state, and zip code.

Product info must include name, description, color, and size.

Each product sold or for sale must be identifiable, so the E-R must be able to store the sale date if sold, and the current status of the sales process (available, ordered, shipped, delivered).

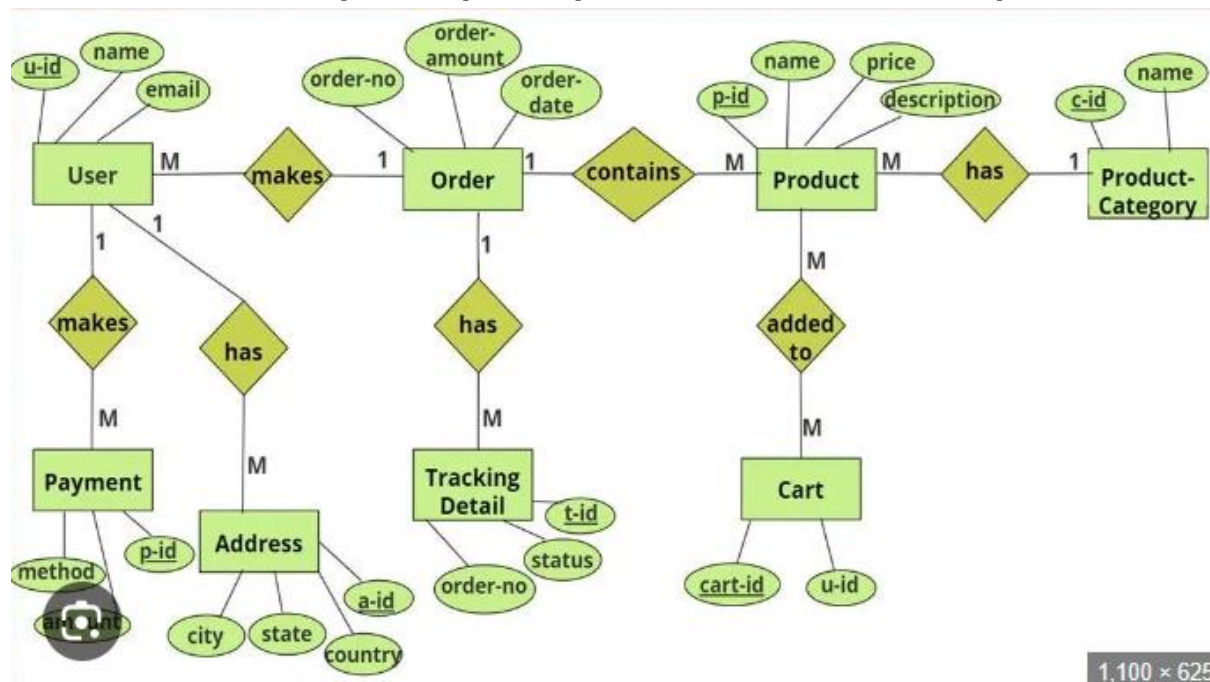
Develop an ER diagram for each of these questions based on Ebay.

1. Define entity sets that contain all data objects that you need.

Define attributes and a primary key for each entity set

2. Define relationships between entities that reflect the requirements completely

3. Define total and partial participation in the relationships



an approximated answer

14.solve the question

Karl Andersson has a boat rental. As summer season approaches, he thinks of the mess of paper that he has to deal with every year. It would be so much more convenient to have everything organized digitally. His friend comes up with a first draft of an ER model shown in Figure 1.

Figure 1. ER model

Every boat has a license plate that identifies the boat, a name, and a type (sailing-, motor boat etc.). Furthermore, Karl stores the customer's name and address (pcode is the postal code). Every rental has a number. A boat is rented from a certain date and time to a certain date and time. The status stores the information whether the boat has been returned or not.

(a) Mark the primary key in the ER model (adjust the model if necessary) and give a short explanation, why the primary keys are possible to use. (2 pts)

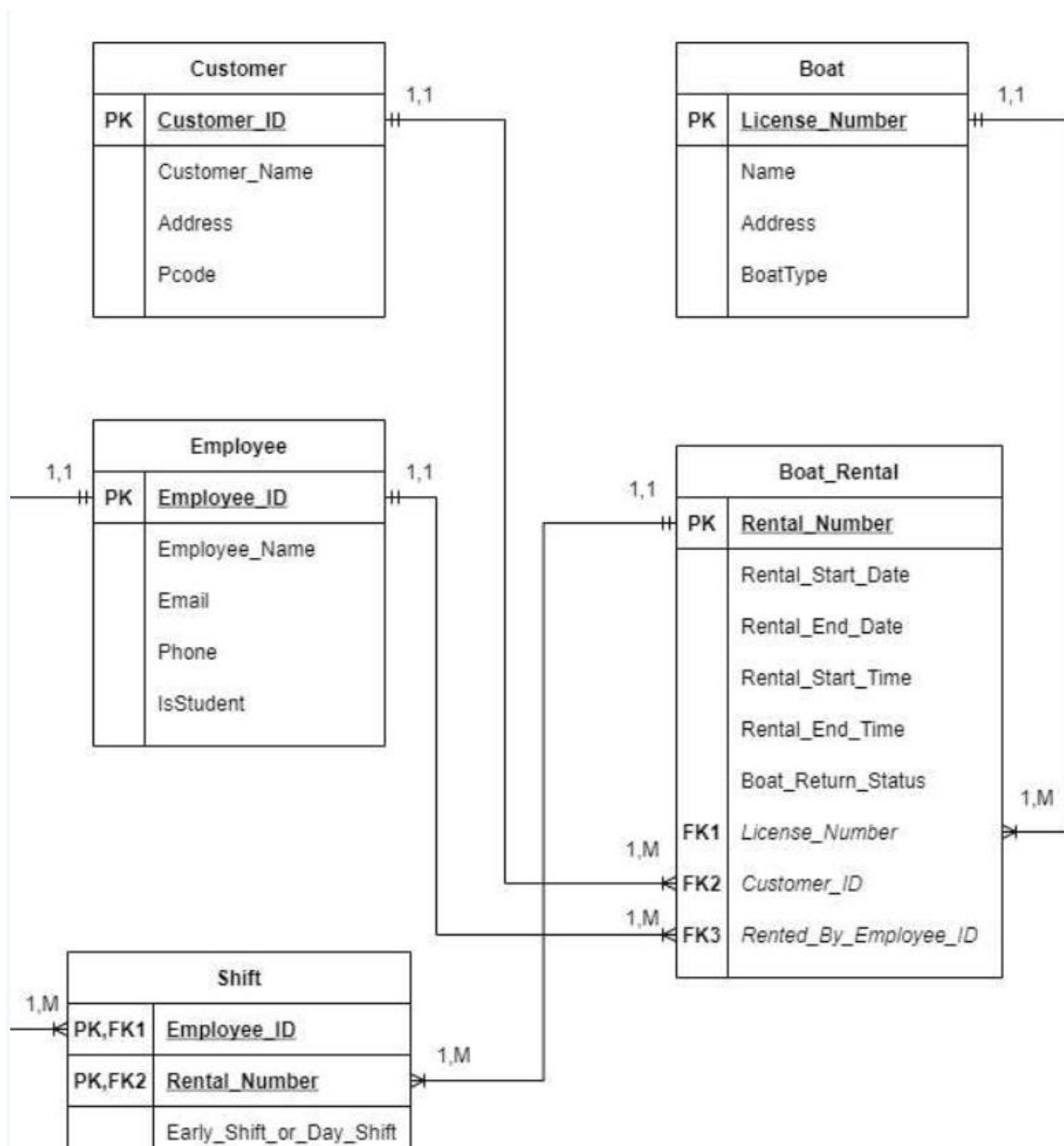
(b) Add cardinality ratios and briefly explain your decision. (3 pts)

(c) Translate the ER model to the relational model. For the relationship types: Do you need an extra table? Why or why not? (5 pts)

d) Extended ER diagram for the given scenario

Relationships of cardinality ratios

- Each customer can rent a boat at least as one to one (1:1) in minimum cardinality ratio while a customer can rent one to many boats on the same day at different times on different dates on different times as one to many (1:M) in maximum ratio.
- Each boat is involved in one rental at a time as one to one (1:1) in minimum cardinality ratio while a boat can be involved in many rentals on the same day at different times or on different dates and at different times as one to many (1:M) in maximum ratio.
- A student employee can do a rental as one to one (1:1) in minimum cardinality ratio while a student employee can do many rentals on the same date or on different dates and different times as one to many (1:M) in maximum ratio.
- Each student is scheduled for one shift while more than one student can be scheduled for one shift as one to one (1:1) in minimum cardinality ratio as one to many (1:M) in maximum ratio.
- Since more than one student is scheduled for one boat rental shift, the entity *Shift* needs to be a weak entity for which a composite primary key has to be created as **partial keys** and also act as foreign keys.
 - Employee_ID is a partial key as it inherits the primary key of the entity Employee
 - Rental_Number is a partial key as it inherits the primary key of the entity Rental.



15. Given the relational schema.

Driver(Name, LicenseNumber, Address)

Car(LicensePlate, Type, Manufacturer)

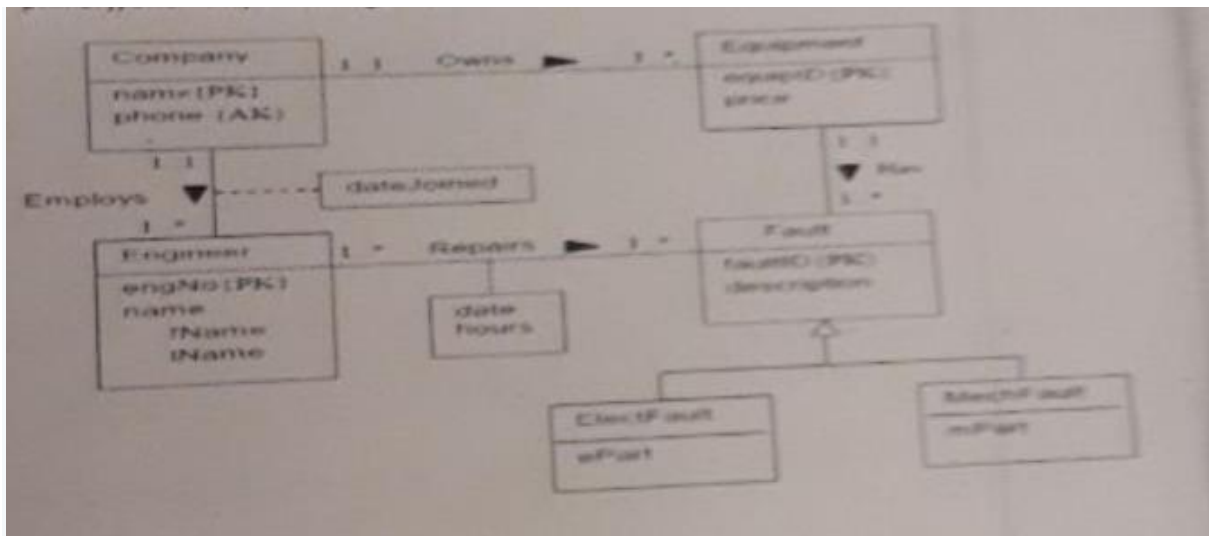
Owner(Licenseplate, LicenseNumber, Year_of_purchase)

Write a relational algebra expression that gives all the drivers living in Debrecen and have a Lexus car purchased in 2013 [For half mark you can write just an SQL expression.]

solution:

π Name(σ Address='Debrecen' &Type='Lexus'
&Year_of_purchase=2013 (Driver $\bowtie$ Owner $\bowtie$ Car))

16. Provide a set of relational tables for the high level data model shown below .Identify primary , alternate and foreign key in the table.



a)

Below is the relational schema of the given high level data model.

It consist of 8 relations. Each relation contains a primary key (mentioned as underlined) and possible foreign key (as bold).

Relation fault has specilization relations ElectFault and MechFault.

Company (Name, phone)

Equipment (equipID, price)

CompanyEquip (Name fk, equipID, dateOfPurchase)

Engineer (engNo, fName, lName, Name fk, dateJoined)

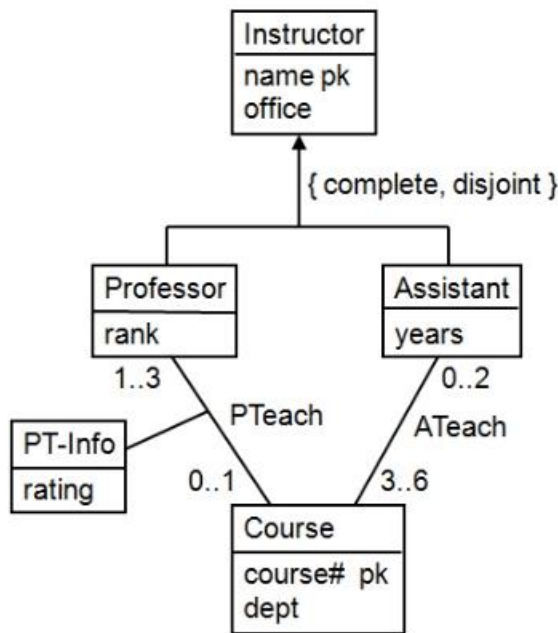
Fault (faultID, description, equipID fk)

Repair (engNo, faultID, date, hours)

ElectFault (faultID, ePart)

MechFault (faultID, mPart)

17. Provide a set of relational tables for the high-level data model shown below. Identify primary and foreign keys in the tables. There are several possible automatic translations; use the translation for subclassing most appropriate for the specified properties. (8 points)



Professor(name,office,rank,course#,rating)

Assistant(name,office,years)

Course(course#,dept)

ATeach(name,course#)

18. a) In the database context, what do we mean by redundant data? (1 point)

b) Why might it be a good idea to have redundant data in a database? (2 points)

c) Why might it be a bad idea to have redundant data in a database? (2 points)

a) Redundant data in a database refers to the repetition of the same data in multiple places. This duplication can occur within a single table or across multiple tables, leading to multiple copies of the same piece of information.

b) Data Availability: Redundancy can increase data availability and

reliability. If one instance of the data becomes corrupted or unavailable, other copies can still be used, enhancing fault tolerance.

c) **Data Inconsistency:** Redundant data can lead to data inconsistency. If one copy of the data is updated while others are not, it can result in conflicting information within the database. This makes maintaining data integrity more challenging.

19. Consider the following relation for published books:

BOOK (Book_title, Author_name, Book_type, List_price, Author_affil, Publisher)

Author_affil refers to the affiliation of author. Suppose the following dependencies exist:

Book_title $\rightarrow$ Publisher, Book_type

Book_type $\rightarrow$ List_price

Author_name $\rightarrow$ Author_affil

a) What normal form is the relation in? Explain your answer. (2 points)

b) Apply normalization until you cannot decompose the relations further. State the reasons behind each decomposition. (4 points)

Given relation is

BOOK(Book_title, Author_name, Book_type, List_price, Author_affil, Publisher)

Given functional dependencies are:

Book_title $\rightarrow$ Publisher, Book_type

Book_type $\rightarrow$ List_price

Author_name $\rightarrow$ Author_affil

a.

- The given relation is in 1NF (1st Normal Form)
- The keys here are Book_title and Author_name
- As no fully functional dependent attributes exists on the key, it is not in 2NF (2nd Normal Form)
- It is not in 3NF (3rd Normal Form) because the keys are partially functional dependent.

b.

- Decompose to 2NF:
- Book_0 (Book_title, Author_name) => Book_title, Author_name is key
- Book_1 (Book_title, Publisher, Book_type, List_price) => Book_title is key
- Book_2 (Author_name, Author_affil) => Author_name is key
- Now the relation is in 2NF as it eliminates the partial dependencies.

- Decompose to 3NF:
- Reasons for the relationship not in 3NF:
- Book_title -> Book_type -> List_price
- Book_type is not a key as well not a subset of a key
- List_price is not a prime attributes
- Decomposition:
- Book_0 (Book_title, Author_name)
- Book_11 (Book_title, Publisher, Book_type)
- Book_12 (Book_type, List_price)

- Book_2 (Author_name, Author_affil)

- Now the transitive dependencies on List_price is eliminated

20. Consider the relation REFRIG (Model#, Year, Price, Manuf_planet, Color), which is abbreviated as REFRIG (M, Y, P, Mp, C), and the following set F of functional dependencies:

$F = \{ M \rightarrow Mp, \{ M, Y \} \rightarrow P, Mp \rightarrow C \}$

a) Evaluate each of the following as a candidate key for REFRIG, giving reasons why it can or cannot be a key: {M}, {M, Y}, {M, C}.

b) Based on the above key determination, state whether the relation REFRIG is in 3NF and in BCNF, giving proper reasons.

c) Consider the decomposition of REFRIG into:

$D = \{R1 (M, Y, P); R2 (M, MP, C)\}.$

Is this decomposition lossless? Show why!

Answer:-----

a.

i) {M} is not a candidate key since it does not functionally determine attributes Y or P.

ii) {M, Y} is a candidate key since it functionally determines the remaining attributes P, MP, and C. From F it is clear that MP and P are functionally determined by M and Y. By transitivity ($M \rightarrow MP$ and $MP \rightarrow C$) we also know that $M \rightarrow C$.

iii) {M, C} is not a candidate key since it does not functionally determine attributes Y or P.

b.

BCNF can be directly tested by using all of the given dependencies and finding out if the left hand side of each is a super key. Take the two FD's $M \rightarrow MP$ and $MP \rightarrow C$, since neither M nor MP is a super key, we can conclude that REFRIG is neither in 3NF nor in BCNF. Alternatively, REFRIG is not in 2NF, because {M,Y} is a key. So, we can see here, Partial Functional dependency exists i.e ($M \rightarrow MP$).

So it is not also in 2NF,
Therefore, Given relation is in 1NF.

c).

Given the decomposition of REFRIG into D:

$R_1 (M, Y, P)$

$R_2 (M, MP, C)$

Using the test for Binary Decomposition, we calculate:

$(R_1 \cap R_2) = \{M\}$; $(R_1 - R_2) = \{Y, P\}$; $(R_2 - R_1) = \{MP, C\}$

Then, we have: $\{M\} \rightarrow \{MP\}$ (given) $\{M\} \rightarrow \{C\}$

$\{M\} \rightarrow \{MP\}$ and $\{MP\} \{C\}$ $\{M\} \rightarrow \{MP, C\}$

So, $(R_1 \cap R_2) \rightarrow ((R_2 - R_1))$ or the decomposition D is lossless.

21.solve the question

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4) Consider the relation Employee_Department in the following figure.

Enum	Bdate	Dnumber	Department	Dhomepage
John B	1965-01-03	2	Computing Science	www/it/cos
Smith L	1955-10-12	2	Computing Science	www/it/cos
James P	1962-03-03	1	Computer Systems	www/it/sys
Anna R	1976-11-23	4	Scientific Computing	www/it/sci
Pia S	1964-10-10	4	Scientific Computing	www/it/sci

The relation contains the following attributes:

Enum: Employee name, Enum: Employee number, Bdate: Employees birthdate, Dnumber: Department number, Dhomepage: Department homepage.

- What update anomalies may occur in Employee_Department? (2 points)
- List all the functional dependencies we have in this table! (2 points)
- In which normal form (NF) is the relation Employee_Department and why? (2 points)
- Show how you would normalise this table into 3NF. (4 points)
- Sketch an Entity – Relationship model for the data shown in Table. Show all the entities, relationships, and attributes!! (3 points)

a) If we update the Dhomepage of computer science dept. we need to update in two tuples, otherwise the data will be inconsistent.

b) {Enum} → {Enum, Bdate}

{Dnumber} → {Department, Dhomepage}

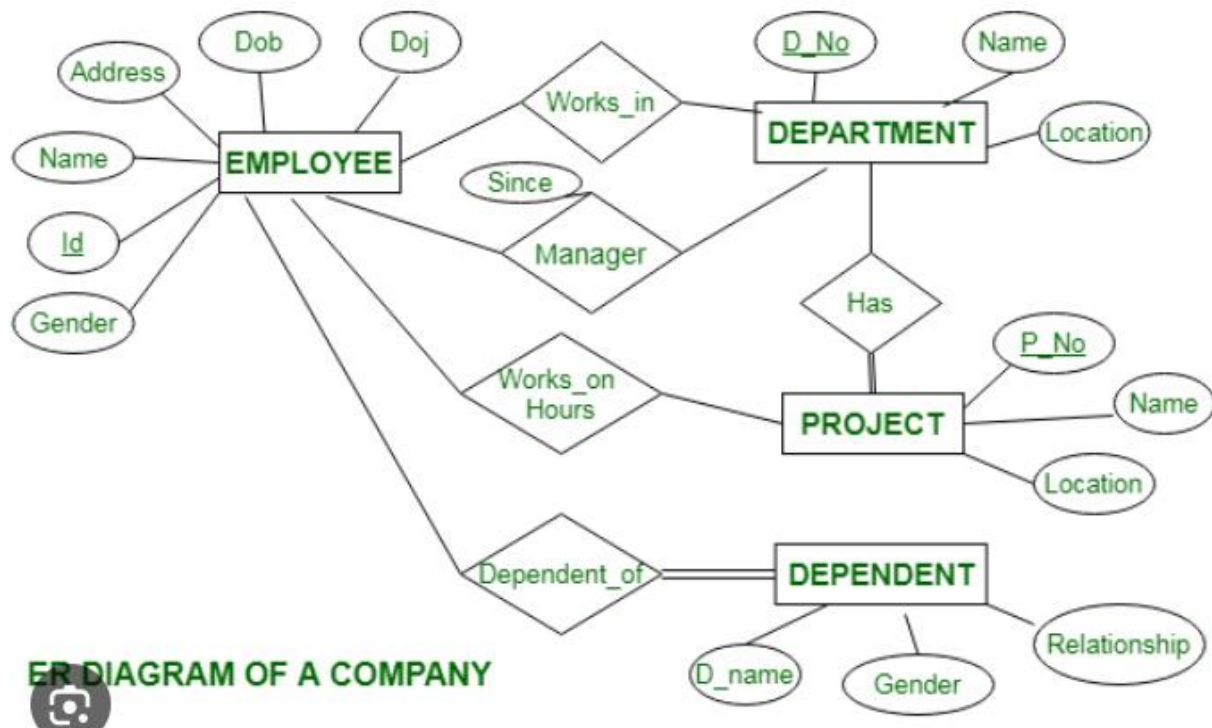
c) It is in 1NF because all attributes are simple and atomic

not in 2NF because PK(Enum, Dnumber) where Enum → Enum so partial dependency

d) Employee(Enum, Enum, Bdate, Dnumber*)

Department(Dnumber, Department, Dhomepage)

e)



22. The following table stores details of doctors, patients and dates of appointments. The Primary Key is (doctorID, patientID).

<u>doctorID</u>	doctorName	<u>patientID</u>	patientName	date
D01	Kumar	P02	Smith	10-August
D01	Kumar	P01	Ford	15-September
D02	Robinson	P02	Smith	10-August

a. Explain why this table is not in 2NF. (2 points)

b. Describe two types of anomalies that could be caused by update, insert or delete operations giving an example of each, with reference to the table above. (4 points)

c. Transform table into 2NF. Show the structures of the resultant tables. (4 points)

a. The table is not in 2NF because here primary key is {doctorID, patientID} where doctorID

-> doctorName. So here is partial dependency

b. Deletion Anamoly: If we want to delete the appointment date of Smith the doctors detail will also be lost

insert anomaly: If we want to insert a new doctor it will not be possible unless he has an appointment

c.Doctor(doctorID pk,doctorName)

Patient(patientID pk, patientName)

Appointment(doctorID,patientID,date)

23.Some Extra MCQs

QUESTION 3..Entity A is a _____ as it can not exist in the database unless another type of entity also exist in the database

- A. Weak Entity.**
- B. Strong Entity.**
- C. Entity.**
- D. None of the given.**

Ans:-- Option A. (Weak Entity).

Explanation:--Weak entity cannot exist in the database unless another type of entity also exists in the database, but does not require that the identifier of that other entity be included as part of its own identifier.The identifier of the weak entity must be a composite key.

QUESTION 4...A recursive relationship is a relationship between an entity and_____.

- A. Strong Entity.**
- B. Weak Entity.**
- C. Composite Entity.**
- D. Itself.**

Ans:---Option D. (Itself).

Explanation:--A recursive relationship is a relationship between an entity and itself. In relational databases, when a table is involved in a relationship with itself, it is called a recursive relationship. For example, in a supervisor-supervisee relationship, a table storing employee records is involved in a relationship with itself.

A strong entity is an entity type whose existence doesn't depend on any other entity.

A weak entity is an entity that cannot be uniquely identified by its attributes alone; therefore, it must use a foreign key in conjunction with its attributes to create a primary key.

Composite Entity.:-This entity lies between the two entities that are of interest and this composite entity shares the primary keys from both the connecting tables.

QUESTION 5:--Which of the following is Not True regarding Weak Entity?

- A. None of the given.**
- B. Identifying relationship.**
- C. Primary key derived from parent entity.**
- D. Has an Existence dependency.**

Ans:--Option:--A..(None of the given).

Explanation:---The weak entities have total participation constraint (existence dependency) in its identifying relationship with owner identity.

A weak entity has a primary key that contains only one foreign key. The entity DEPENDENT, for example, is a weak entity because it uses the primary key of the EMPLOYEE entity as part of its own primary key.

In general (though not necessarily) a weak entity does not have any items in its primary key other than its inherited primary key and a sequence number.

The Relationship that connects the weak entity to its owner identity is called Total Participation in the identifying relationship.

QUESTION 6...Which of the following is True regarding Associative entity ?

- A. Called bridge entity**
- B. Primary key is a composite attribute**
- C. Used to resolve complexity in many to many relationship**
- D. All of the given**

Ans:--Option D. (All of the Given).

Explanation:---

Associative entities are implemented in a database structure using associative tables, which are tables that can contain references to columns from the same or different database tables within the same database.

Associative tables are colloquially known under many names, including association table, bridge table,

. A relational database requires the implementation of a base relation (or base table) to resolve many-to-many relationships. A base relation representing this kind of entity is called, informally, an associative table.

Associative entities don't have a primary key based on their own attributes. An associative entity has the same primary key as its supertype associative entity, and this primary key is also a foreign key to the supertype associative entity. An associative (or junction) table maps two or more tables together by referencing the primary keys (PK) of each data table. In effect, it contains a number of foreign keys (FK), each in a many-to-one relationship from the junction table to the individual data tables. The PK of the associative table is typically composed of the FK columns themselves.

QUESTION 7....The attribute AGE is calculated from DATE OF BIRTH. The attribute AGE is:

- A. Derived Attribute**
- B. Stored Attribute**
- C. Multivalued Attribute**
- D. Composite Attribute**

Ans:---Option A.. (Derived Attribute).

Explanation:--The value for this type of attribute can be derived from the values of other related attributes or entities.

A stored attribute is an attribute that cannot be derived from other attributes.

A multivalued attribute of an entity is an attribute that can have more than one value associated with the key of the entity.

The attributes which can be divided into sub-parts are called composite attributes

QUESTION 8...An attribute that identify an entity is called

- A. Entity.**
- B. Relationship.**
- C. Composite Key.**
- D. Identifier.**

Ans:--Option D.. (Identifier).

Explanation:--Entity identifiers are attributes, specifically, key attributes that uniquely identify each entity.

An entity in DBMS (Database management System) is a real-world thing or a real-world object which is distinguishable from other objects in the real world.

A composite key is a candidate key that consists of two or more attributes (table columns) that together uniquely identify an entity occurrence (table row).

Relationships allow relational databases to split and store data in different tables, while linking disparate data items.

QUESTION 9..Which of the following indicate the maximum number of records that can be involved in a relationship?

- A. Maximum Cardinality.**
- B. Maximum Connectivity.**
- C. Minimum Connectivity.**
- D. Minimum Cardinality.**

Ans:--Option A...(Maximum Cardinality.).

Explanation:--Maximum cardinality is defined as the maximum number of entity classes which are taking part in a relationship. There are three types of Maximum Cardinality: one-to-one, one-to-many, many-to-many. In 1:n , 1 is the minimum cardinality, n is the maximum cardinality.

QUESTION 10,,In which of the following can one entity instance of one type be related to many entity instances of another type?

- A. One to Many relationship.**
- B. Many to Many relationship.**
- C.Zero to Many relationship.**
- D.One to One relationship.**

Ans:--Option A...(One to Many relationship.)

Explanation:--A one-to-many relationship is a type of cardinality that refers to the relationship between two entities A and B in which an element of A may be linked to many elements of B, but a member of B is linked to only one element of A..

A many-to-many relationship is a type of cardinality that refers to the relationship between two entities, say, A and B, where A may contain a parent instance for which there are many children in B and vice versa.

A zero to many optional relationship indicates that a person may have no phone, one phone, or many phones, and that the phone may not be "owned," but can only be owned by a maximum of one person.

A one-to-one relationship is a type of cardinality that refers to the relationship between two entities A and B in which one element of A may only be linked to one element of B, and vice versa.

